

Research Notebook: Evaluation of Time Reduction Rate

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ver. 1.3.0.†

Abstract

Prior to publication, this document was created for the purpose of organizing thoughts, and serves as a log notebook recording collected information, obtained results and findings, and log data. Part of it may potentially become supplementary material for a published paper.

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†Data is available at https://github.com/pub5023/productivity_indicator

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0 Background: History of Measurement Experiments

The time reduction rate is one of the metrics used to express how much business operations have improved through the use of generative AI. Several validation reports on this topic have been published[1]-[7]. There are few examples of experimental evaluation of the actual reduction effect achieved by using AI in real business settings. Since these results are valuable as reference ground-truth data when having AI estimate reduction rates, the contents are reviewed below.

ref.[1] This study compared a “test group” using a GPT-3.5-based internal tool developed by Trane Technologies against a “control group” performing tasks manually, and measured the time reduction rate for key tasks related to document creation.

Task-type differences: For tasks involving large amounts of text processing or high working memory load (e.g., summarization), AI-driven productivity gains were dramatic (69%). In contrast, for relatively simpler tasks (e.g., email drafting), the time required to write prompts and review/edit AI responses meant that significant time savings compared to manual work were not achieved (3.3%).

Time reduction rates for the three primary tasks: Article summarization: 69%, instruction writing: 45%, presentation creation: 25%

ref.[2] This study examined “mid-level professional writing tasks” within specific occupations. Based on the period of the study, the ChatGPT version used is considered to be GPT-3.5. (1) Participants: 444 experienced professionals (college graduates) (2) Target occupations: marketing professionals, grant writers, consultants, data analysts, HR specialists, and managers. (3) Specific task examples: writing press releases, producing short reports, drafting analysis plans, composing delicate emails, etc.

Result: While the control group took an average of 27 minutes, the AI-assisted group completed tasks in an average of 17 minutes, achieving a time reduction of approximately 37% (10 minutes): $(27 - 17)/27 = 0.37$.

ref.[3] This study examined the “Corporate Financial Expense Processing” workflow at a major Korean company (Company S). The process consists of: (1). Receipt processing: reading corporate card receipts (paper or image-based unstructured data) submitted by employees. (2). Information extraction: extracting key information such as date, amount, and item using IDP (Intelligent Document Processing). (3). Policy-based classification: determining whether extracted items comply with internal regulations (whitelist/blacklist) and identifying the appropriate account category. (4). Exception handling (critical): for

new products or ambiguous items, using generative AI (LLM) to provide appropriate judgments and recommendations based on internal policy documents.

Result: Overall processing time was reduced by 80%–90%. This improvement goes beyond simple routine-task automation (RPA); a major contributing factor is that generative AI can now read policy documents and present reasoning and recommended account categories for “exceptional cases (e.g., unregistered items)” that previously required human judgment. The use of GPT-4o-mini, which offers improved performance over GPT-3.5, played a significant role. (Samsung SDS’s automation platform “Brity Automation” is used, and within its “AI Flow” feature, the OpenAI API (GPT-4o mini) is called to perform expense item classification and exception determination.)

ref.[4] Unlike conventional controlled laboratory tests, this report analyzed 100,000 conversation logs between real users and Claude to estimate how much AI reduced task completion time. Claude itself calculated estimates by comparing “the estimated time a human would take to perform the task manually” with “the time spent interacting with AI”: According to an internal productivity analysis by Anthropic, analysis of 100,000 real-world AI-assisted task interactions suggests that task completion time was reduced by an estimated around 80% on average when assisted by AI, based on anonymized usage logs (Anthropic, 2025).[4]

Summary of results (tasks by reduction rate):

(1). Tasks with high effectiveness (90%+ reduction)

Tasks where AI’s “reading/writing speed” and “information organization capability” produce dramatic effects.

- Educational curriculum creation (96%): vocational education teachers creating lesson plans and course content.

- Report information editing (95%): extracting and integrating information from multiple reports.

- Research result summarization (91%): social science research assistants creating tables, graphs, and fact sheets, and summarizing results.

(2). Tasks with moderate to high effectiveness (70–89% reduction)

Includes general business document creation and analytical tasks.

- Administrative document creation (87%): secretarial tasks such as creating invoices, reports, memos, and financial statements.

- Financial analysis (80%): analysts interpreting data such as prices, yields, and investment risks.

- Student advising (83%): agricultural teachers advising students on career paths and curricula.

(3). Tasks with relatively low effectiveness (60% or less reduction)

Tasks requiring physical verification or that experts can already perform quickly show limited effects.

- Office equipment troubleshooting (56%): resolving computer hardware and software issues.

Regarding trends by occupation, high time reduction effects were observed in complex knowledge work with higher wages, such as management (85.1%), business/finance (83.5%), legal (80.7%), and engineering (77.7%). In contrast, sales (66.2%) and installation/repair (65.7%) showed relatively lower results.

ref.[6] This study is one of the earliest important large-scale field validations of the economic impact of generative AI in actual workplace settings. Task evaluated: customer support (technical support) division of a software company serving Fortune 500 clients. Result: 9% reduction in per-case handling time. 14% reduction in multi-chat simultaneous handling capacity. Based on the period of the study, GPT-3 is estimated to have been used.

ref.[7] Using a virtual assistant for taxpayers (Inland Revenue Authority of Singapore — IRAS), automated responses to general taxpayer inquiries (“AskJamie” and subsequent GenAI adoption). Response time was reduced by up to 80%.

1 Construction of Occupation Ground Truth

Abstracting the three primary tasks evaluated in ref.[1], the central work involves handling text, organizing information, and restructuring it into a form that can be communicated to others. (Three primary tasks: document summarization T_1 , instruction writing T_2 , presentation creation T_3)

Typical occupations with these characteristics include:

1. Consulting

Examples: management consultant, IT consultant, business improvement consultant, strategy analyst.

Characteristics: Daily tasks consist of summarizing client information, creating instruction documents (to-be processes, requirements definitions), and producing presentation materials.

2. Planning and marketing roles

Examples: planning (business/product planning), marketing, public relations/PR.

Characteristics: Frequent production of internal and external materials. Summarizing market research, creating planning documents and instruction sheets, and preparing internal presentation materials are essential skills.

3. Project Manager (PM) / Product Manager (PdM)
Examples: IT project manager, product manager.
Characteristics: Extensive document creation to drive projects. Large volumes of requirements documents, task instruction sheets, and stakeholder presentation materials are produced.
4. Editors, writers, and researchers
Examples: editors (publishing/web media), researchers, technical writers.
Characteristics: Primarily focused on text summarization and structuring. Daily tasks include article summarization, structural planning, and documentation.
5. Clerical and assistant roles
Examples: sales assistants, secretaries, general affairs/administration departments.
Characteristics: Frequent organization and creation of internal documents. Creating meeting minutes, summaries, instruction sheets, and internal briefing materials.
6. Education and training roles
Examples: training instructors, instructional designers, corporate trainers.
Characteristics: Primarily focused on material summarization, manual creation, and training presentation preparation.

Generalizing the subjects evaluated in ref.[2], the above occupation groups 1, 2, 3, and 4 apply. The flow structure of summarizing acquired information (T_1), processing based on conditional rules and formats (T_2), and reporting/presenting results (T_3) follows a common business framework of input, processing, and output, even beyond document-creation-specific occupations. Looking at additional occupations from other literature beyond the above, there are finance/legal occupations[3, 4] and customer support occupations[4, 6, 7]. Their recorded performance values and task characteristics are organized below.

Beyond ref.[3] (80–90%), recorded performance values for finance/accounting include financial analysis (80%), business/finance (83.5%), and legal (80.7%) from ref.[4]. In this paper, we use 80% as the average.

Recorded performance values for customer support: 80% in ref.[7] (taxpayer response), and office equipment troubleshooting (56%) in ref.[4]. (ref.[6] (9%) uses the older GPT-3, so it is excluded.)

7. Finance and accounting occupations
Characteristics: Daily tasks involve reviewing large volumes of employee-submitted receipts, verifying compliance with internal regulations (policies), classifying them,

and approving/registering them in accounting systems (ERP).

Examples: financial manager, accounting staff, expense reimbursement administrator, corporate finance officer

Primary tasks:

T_1 : Summarizing inquiries/logs and organizing key issues (receipt processing and information extraction)

T_2 : Reviewing policy compliance and classifying account categories (regulation checking and journaling)

T_3 : Exception handling and final approval (decision-making and system registration)

8. Customer support and technical support occupations

Characteristics: Daily tasks involve diagnosing technical inquiries from customers and providing real-time solutions through chat systems. Examples: technical support agent, helpdesk staff, customer success manager, chat support operator

Primary tasks:

T_1 : Customer problem diagnosis and information gathering (inquiry and situation assessment)

T_2 : Proposing solutions and composing responses (solution delivery)

T_3 : Knowledge accumulation and learning (organizational knowledge sharing)

1.1 Occupation Descriptions and Task Attributes

For the eight occupation groups above, “occupation descriptions” are prepared that describe an overview of each job, examples of relevant occupations, and an overview and specific details of the three primary tasks.

Even for the same task content, its nature differs by occupation; to characterize this, the concepts of task time share weight w_t and importance W_t are introduced for task t . Since occupations involve a variety of tasks, obtaining comparable time share ratios is difficult, so in this paper, these are defined by the time ratios of the three primary tasks. Specifically, with reference to the occupation descriptions, generative AI is used to estimate the time ratio (composition ratio of three tasks) for human task execution and the task importance for each of the three primary tasks (Table 23).

As noted in ref.[1], the reduction effect varies with the nature of the task, and reduction rates also vary with the skill of the person performing the task[1, 2]. To partially account for these circumstances, occupation descriptions with derived variants of task content are also prepared for some occupations, and their

evaluations are included (those with branch number 2 appended to the occupation id: 2.2, 7.2, 8.2). Specifically, for 2.2 (marketing professional), the same occupation description is used, but elements of management planning/business planning are excluded when making estimates. For 7.2 (finance/accounting) and 8.2 (customer support), occupation descriptions focused on document creation for the three primary tasks were created.

ChatGPT 5.1 Instant was used for creating occupation descriptions and estimating importance and time ratios. Although the content of these descriptions and estimated values will differ depending on which AI is used, since these are treated as fixed and given throughout this paper, differences due to AI choice are not considered.

1.2 Benchmark Time Reduction Rate BM/ref4

The ref.1 column of Table 23 contains values calculated by taking the time-weighted average of the three primary task time reduction rates obtained in ref.[1] (summarization 69%, instruction creation 46%, presentation creation 25%) using each occupation’s time ratios. ref.1 and ref.2 are primarily results from GPT-3.5-class models, and for occupations 1–4, the reduction rate of 37% based on ref.[2] is recorded in the ref.2 column. Many reduction rates fall below 50%, with manual human work accounting for a large proportion. For occupations 7 and 8, the ref.1 value is not recorded; instead, the value of 80% from the literature[3, 7] is adopted in the ref.2 column (finance[3], customer support[7]). These are estimated to be GPT-4o-mini results ([3] states this explicitly), and while the values are somewhat high, they are inferior to the recent GPT-5 series and are treated as a reference substitute for GPT-3.5.

The third column (ref.4) records reduction rates for the corresponding occupations from [4]. Based on the period of the study, ref.4 is considered to reflect LLMs of Claude 3.5 or later (and the LLMs for occupations 7 and 8 are comparable), and since reasoning capability has improved significantly over GPT-3.5, the reduction rates are also higher.

The BM value is the average of ref.1, ref.2, and ref.4 (a two-value average when one is missing). Therefore, BM, which is an average including GPT-3.5, is at a level higher than GPT-3.5 but represents a conservative value. This BM and the ref.4 column serve as the comparison benchmarks for the reduction rates to be estimated hereafter.

2 Introduction: The Productivity Estimation Problem

2.1 Introduction

In deploying generative AI for business operations, what is important — beyond the ability to correctly perform tasks (quality) — is the ability to estimate in advance the degree to which human work will be reduced by that deployment (productivity). However, estimating time reduction rates is more difficult than simple performance evaluation. The reduction rate is not determined solely by “what the AI outputs,” but depends strongly on residual work such as review, approval, stakeholder coordination, exception handling, and rework, as well as on organizational operating conditions (quality standards, risk tolerance, data constraints). Therefore, a high ranking on general LLM ranking metrics (e.g., comprehensive evaluation of dialogue performance)[8, 9] does not necessarily translate directly into accuracy of time reduction rate estimation[10].

This study verifies the degree to which LLMs can reliably estimate the time reduction rate for business tasks, using a cross-occupation common task design, and quantitatively decomposes the underlying factors of estimation error. For 11 occupations, three common tasks (T_1 : summarization, T_2 : instruction writing, T_3 : presentation preparation) are defined, and task-level reduction rates are computed from the “human time” and “AI time” estimated by the model. For reference values (the so-called “ground truth”), experiment-based benchmarks (BM) are adopted, while giving emphasis to error definition and robustness evaluation in light of the fact that the reference values themselves are condition-dependent quantities.

The main contributions of this study are the following three points. **First**, we show that estimation error is strongly influenced not only by model capability but also by differences in “Estimation Style,” and that pooling all models in a single regression obscures the underlying structure due to cancellation effects. **Second**, even when only occupation-level reference values are available, a simple decomposition (Task-truth decomposition) can estimate task-level reference values, enabling consistent error evaluation at the task level. **Third**, proxy variables extracted from model rationale text (residual work, scope broadening, cautious stance) are shown to be consistent with more conservative estimation (suppression of over-estimation), and the robustness of this conclusion is validated through sensitivity analysis under missingness.

2.2 Related Work

2.2.1 LLM evaluation platforms/leaderboards

Research on evaluating LLMs is growing, and benchmark suites and leaderboards that aggregate cross-task performance are widely used[8, 9]. While these metrics are useful for characterizing a model’s general capability, they often show only a weak association with outcomes in actual enterprise deployments. This is because real-world performance depends strongly on factors such as workflow integration method, constraints, experience differences, and human oversight[1, 6]. Recent discussions on evaluation have emphasized the need to align evaluation criteria with metrics that actually matter in decision-making — such as cost, latency, safety, and productivity — rather than treating a single benchmark as an all-purpose utility indicator[10].

2.2.2 Productivity measurement and human-in-the-loop workflows

In research on AI productivity[1, 2, 6, 7], it has been pointed out that the effects actually obtained are influenced not only by residual human tasks such as verification, approval, exception handling, and rework, but also by organizational factors such as training, tool environment, and quality standards[2, 6]. Empirical workplace studies have consistently shown that the measured degree of time reduction varies with task complexity and process design, and that faster document drafting brings structural changes to the task itself — not necessarily equivalent to end-to-end speed improvement[2]. These observations support the view that productivity should be modeled as a condition-dependent quantity rather than a fixed “true value.”

2.2.3 Rationale, verbosity, and interpretable proxy indicators

LLMs often generate reasoning and explanations, which can serve as cues that surface assumptions about uncertainty, risk, and the need for human verification. However, the quality and verbosity of rationale vary considerably across models and configurations, and missing rationale can occur systematically (e.g., older models tend to produce shorter outputs). Prior work has cautioned against treating rationale as causal evidence[11], while supporting its use as an interpretable signal when combined with robustness checks[12]. Our proxy approach follows this line of thinking, using lightweight and transparent lexical proxy indicators and verifying robustness even under missingness.

‡ **Remark 2.1.** Supplementary note on settings

In the validation conducted hereafter, generative AI is used to estimate two quantities — the time required for a human to perform a task and the time required for AI to perform the same task — each with accompanying reasoning. Since the unit of time differs depending on the volume (unit) of the task being estimated, comparison is performed using the time reduction rate (the ratio of the two estimates). Reasoning is requested in order to record how the model defines the scope of AI involvement, since estimated time varies accordingly. Mixing estimates with different estimation policies reduces analytical precision, so reasoning is used to enable higher-resolution analysis.

Even when the same reference ground truth (“true value”) is used, the meaning of the true value is not the same across experiments, since the capability of the subject population, the scope of the task, and task-specific constraints all differ; there is a latent error that does not appear in measurements. Therefore, the analysis proceeds on the premise that both AI estimates and ground-truth values carry a certain degree of variability.

Rather than simply performing regression analysis and tracking goodness-of-fit alone, there is sufficient value in an approach that investigates the sources of improvement in fit to estimate which variables contribute to variability in estimates.

Evaluation is performed using the deviation ε_i of the estimate from the ground truth and the divergence rate χ_i :

$$\varepsilon_i = \text{estimate} - \text{ground truth} \quad (2.1)$$

$$\chi_i = (\text{estimate} - \text{ground truth}) / \text{ground truth} \quad (2.2)$$

The subscript $i = 1, 2$ denotes values relative to ground truth BM/ref.4 respectively. In the main text, subscripts are omitted to avoid notational clutter.

‡ **Remark 2.2.** Rationale for model selection and evaluation metrics

Since there is expected to be a correlation between the AI capability at the time of the experiments and the ground truth values, in addition to the models GPT-3.5 and GPT-4o-mini used in ref.1 and ref.2, recent reasoning-class models from the GPT series are adopted. For ref.4, the Claude series (Haiku/Sonnet/Opus) is adopted. The inclusion of the latest models is based on the expectation that AI with improved capability would also produce more accurate estimates. There is also a research interest in whether improvements in capability translate into more accurate time estimation.

Furthermore, to ensure variety in explanatory variables for regression analysis, models with lower reasoning capability also need to be included in the analysis. Table 24 lists all models

used. From Chatbot Arena <https://openlm.ai/chatbot-arena/> (accessed: Mar. 4, 2026, JST 19:59), models for which Elo rating (Arena Elo), AAI, and MMLU-Pro values all existed were extracted, and correspondence with the models used was established. Where no exact model name match exists, the closest approximation was selected.

3 Problem Setting and Data

3.1 Task Setting

We construct a dataset covering 11 occupations and three common task types T_t to study the estimation of time reduction in knowledge work. The common task types T_t are as follows:

T_1 : Summarization (information intake and integration)

T_2 : Instruction writing (creating specifications/guidelines/instructions)

T_3 : Presentation preparation (creating slides/storylines)

Where needed, these are expressed concretely as $t = \{\text{summary, instruction, presentation}\}$. Each occupation instantiates these common task types in a job-specific context, but the task categories themselves are fixed, enabling cross-occupation comparison.

3.2 Model Output and Time Reduction Rate Estimation

For each (model $\times$ occupation) pair, the following model estimates are collected: Estimated human work time $\hat{t}_{task}^{human}$ (baseline completion time) Estimated AI-assisted work time $\hat{t}_{task}^{AI}$ (time when using the model) From these two estimates, the estimated time reduction rate $\hat{r}_t$ per task is computed as:

$$\hat{r}_t = 1 - \frac{\hat{t}_{task}^{AI}}{\hat{t}_{task}^{human}} \quad (3.1)$$

We analyze the error of this reduction rate $\hat{r}_t$ against two types of reference values. One is the occupation-based observed value (occupation-truth) constructed in §1 above (see C for details), and the other uses task-truth decomposed to the task level (Section 4). Also, based on the analysis detailed in §3.4, models are classified into estimation style segments $S = \{A, B, C\}$:

A : Recent high-performance models (GPT-5.2 Instant/Thinking, Claude Opus/Sonnet/Haiku, etc.)

B : Legacy models

C : Models exhibiting systematically conservative estimation patterns (e.g., tendency to estimate residual work more heavily)

Segment definitions are fixed across all analyses reported in this paper.

3.3 Importance Weights and Task/Occupation Aggregation

We incorporate task importance $W_t \in [0, 1]$ and time-share weights w_t indicating how much each task contributes to the overall workload of an occupation. These weights serve two purposes:

Computing occupation-level aggregate values (time-weighted average)

Verifying whether importance interacts with task type and explains estimation error

‡ **Remark 3.1.** Deviation and divergence rate of time reduction rate

Representative examples of the deviation and divergence rate of time reduction rates by occupation and task are shown in Table 25 and Table 26. Table 25 shows a conservative-estimating model from segment C (GPT-5.1 Thinking), while Table 26 shows an optimistic-estimating model from segment A (Haiku 4.5). The task-level time reduction rate $\hat{r}_t$, the time-share-weighted average thereof

$$\hat{r}_{occ} = \sum_t w_t \hat{r}_t, \quad (3.2)$$

and deviation ε_i and divergence rate χ_i are listed for all occupations.

Tables 27 and 28 incorporate importance weights, and for the same models respectively, show the time-share-weighted average of task importance multiplied by the time reduction rate:

$$\hat{r}_{occ} = \sum_t w_t W_t \hat{r}_t. \quad (3.3)$$

Tables 29 and 30 show cross-occupation aggregations, listing the occupation averages for all models.

Table 29 shows the occupation-averaged mean reduction rates and errors per model. Except for GPT-5.1 and GPT-5.2 Pro, the ref.4 error is smaller for all models. In contrast, Table 30 shows the occupation average when importance is multiplied into the reduction rate; here the BM error is smaller for all models: GPT-5.1 and GPT-5.2 Pro show smaller errors than the ref.4 values in Table 29, and all other models are comparably small to the ref.4 values in Table 29. Therefore, GPT-5.1 and GPT-5.2 Pro may differ from other models in some property.

3.4 Segment Hypothesis

First, we outline the regression relationship between estimation error and model performance metric X_{elo} (Arena Elo) as explanatory variable:

$$Y_{mae} = \beta X_{elo} + \alpha, \quad (3.4)$$

For brevity, when the regression coefficient β and intercept α are omitted, the symbol $\sim$ is used:

$$Y_{mae} \sim X_{elo}. \quad (3.5)$$

The error used as the dependent variable is the mean absolute error Y_{mae} (mean absolute error), obtained by averaging the absolute error $|\varepsilon_{occ}|$ obtained per occupation over $n = 11$ occupations:

$$Y_{mae} = \frac{1}{n} \sum_{occ} |\varepsilon_{occ}|, \quad \varepsilon_{occ} = \hat{r}_{occ} - r_{occ}^* \quad (3.6)$$

where $\hat{r}_{occ}$ is the occupation-based time reduction rate (3.2) or (3.3). Table 1 summarizes the regression coefficients and correlation coefficients of Y_{mae} with respect to the occupation-truth BM/ref4 (B/R) for the time-weighted average (time/imp; time weight is w_t , imp weight is $w_t W_t$) of three tasks. Figure 1 presents representative scatter plots. The combination of truth type and weight type is denoted as $C[B/R, time/imp]$.

It is expected that higher performance metric X_{elo} corresponds to smaller error Y_{mae} , and Figure 1(b) shows this as expected, but as in Figure 1(a), this tendency is not necessarily apparent.

- (a) $C[B, time]$ ($Y_{mae} = time_abs_ae_bm$) shows nearly zero correlation; Pearson $r = +0.031$ ($p=0.913$) / Spearman $r = -0.159$ ($p=0.571$)
- (b) $C[B, imp]$ ($Y_{mae} = imp_abs_ae2_bm$) shows a clear tendency for higher performance metric X_{elo} to correspond to smaller error; Pearson $r = -0.531$ ($p=0.0418$) / Spearman $r = -0.645$ ($p=0.0095$)
- (c) $C[R, time]$ ($Y_{mae} = time_abs_ae_ref4$) A negative correlation is also maintained against the ref4 benchmark (though Spearman is near the significance boundary, $p=0.0871$); Pearson $r = -0.570$ ($p=0.0264$) / Spearman $r = -0.457$ ($p=0.0871$).

Looking at the scatter plot in Figure 1, taking into account differences in the stated estimation reasoning (conservative/optimistic), the models appear to fall into three distinct segments:

A: Recent high-capability reasoning models (GPT-5, Opus/Sonnet/Haiku)

B: Legacy models with sparse reasoning descriptions (all except A and C)

C: Models with conservative reasoning descriptions: gpt52P and gpt51T

This suggests that estimation error is strongly influenced not only by model capability but also by differences in “Estimation

Style,” and that pooling all models in a single regression obscures the underlying structure due to cancellation effects. As in Figure 1(a), fitting a single regression to everything averages out different “estimation policies” and makes them hard to see; this leads to the hypothesis that $A/B/C$ appear to reflect not “capability differences” but rather higher-level differences in “estimation style (calibration policy).”

If this is the case, what is mixed into the regression above includes: structural effects of importance and task (the effect one wants to observe), models’ “bias policy” (conservative/aggressive/legacy), and further compounded mixing from the differences in experimental conditions between the ground truth values (BM/ref.4), making it genuinely difficult to interpret.

(output data) [folder#,subfolder#] filename:

Table 1: [1,1] corr_ArenaElo_vs_errors.csv

Figure 1(a): [1,1] elo_vs_mae_time_bm.png

Figure 1(b): [1,1] elo_vs_mae_imp_bm.png

Figure 1(c): [1,1] elo_vs_mae_imp_ref4.png

3.5 Interaction Effects (Occupation-Truth Reference)

Figure 1(b) suggests that importance plays a role, but let us confirm one more piece of evidence supporting the hypothesis that segment effects are more influential. First, as a baseline, to verify whether the effect of importance on estimation error (over/under-estimation) varies by task type, a regression (OLS, HC3 robust standard errors) is estimated with the task-level signed error ε_t^{occ} as the dependent variable, and importance W_t , task type T_t (summary as the reference category), and their interaction as explanatory variables.

$$\varepsilon_t^{occ} \sim W_t * T_t. \quad (3.7)$$

Since task type participates as an explanatory variable, the error for the dependent variable must also be specified per task type. Here, the signed error is simply defined as the deviation from the occupation-truth; the decomposition into task-truth is described later:

$$\varepsilon_t^{occ} = \hat{r}_t^{occ} - r_{occ}^*. \quad (3.8)$$

The interaction is symbolically expressed as

$$A * B = A + B + A : B \quad (3.9)$$

and concretely, expanding over each component a, b of A, B :

$$A * B = \sum_a \beta_a I(a) + \sum_b \beta_b I(b) + \sum_{a:b} \beta_{a:b} I(a)I(b) \quad (3.10)$$

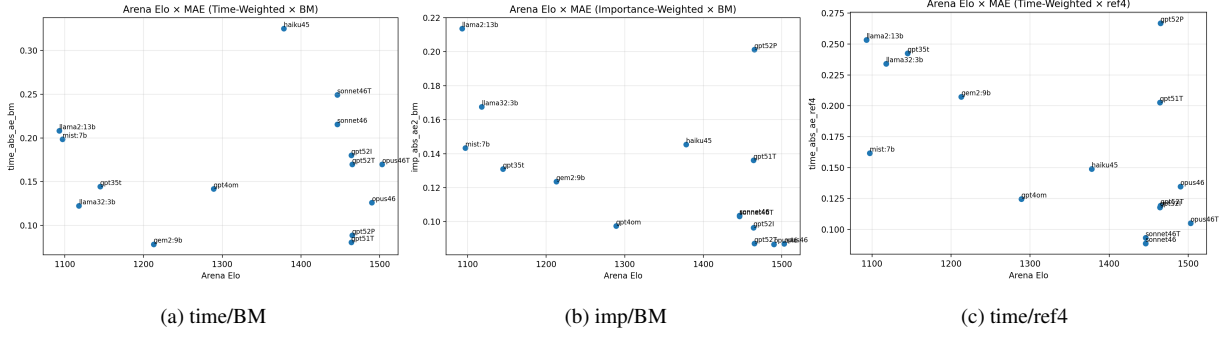


Figure 1: Correlations between Arena Elo and MAE.

Table 1: Correlation between Arena Elo and error metrics

weight	truth type	Pearson r	Pearson p	Spearman r	Spearman p	slope β	slope p
time	BM	0.031	0.913	-0.159	0.5705	1.29867	0.9126
imp	BM	-0.531	0.0418	-0.645	0.0095	-0.00013	0.0418
time	ref4	-0.570	0.0264	-0.457	0.0871	-0.00022	0.0264
imp	ref4	-0.403	0.1366	-0.294	0.2881	-0.00016	0.1366

$a : b$ denotes the cross-term of a and b . When making explicit that a is a category value of nominal variable A , the notation $A.a$ is used. $I(a)$ takes the value 1 in the case of a nominal variable, and the realized value of A otherwise.

The regression coefficients are as shown in Appendix A.1.3 Appendix Table 8; in this case, the interaction between importance and task is not significant at the p-value level for either BM or ref.4. However, the result that the “importance main effect for summary (the reference task) under ref.4” is significant was obtained. Furthermore, looking at the sign of the regression coefficients, there is a tendency for error to shift in the positive direction (toward over-estimation) as importance increases. Under ref.4, this tendency is even stronger, particularly for instruction. This suggests that importance, as a proxy for “ease of implementation/adoption / operational richness,” modifies the pattern of error depending on the task; even at the same importance level, the behavior differs between summary and instruction.

(output data regression coefficient files):

[1,1] regression_coef_importance_task_BM_ref4.csv

[1,1] regression_slopes_by_task_BM_ref4.csv

Here, to verify the possibility that the importance effect on estimation error is contaminated by differences in models’ “estimation style,” the segment term $S = \{s|A,B,C\}$ is explicitly added to (3.7) and re-estimated (analysis 2). It is found that the effect of importance W_t is weak (importance–task interaction,

Appendix A.1.3):

$$\varepsilon_t^{occ} \sim W_t * T_t + S. \quad (3.11)$$

Various segment regressions based on occupation-truth ($r_{occ}^* = r_{occ}^{BM}$) are shown in Appendix A.1, where it is also suggested that explicitly accounting for segment differences is effective in explaining the error structure (Table 10). (Appendix A.1.2)

Baseline regression (Appendix A.1.2)

$$\varepsilon_t^{occ} \sim T_t + S, \quad (3.12)$$

Triple interaction (Appendix A.1.5)

$$\varepsilon_t^{occ} \sim W_t * T_t * S. \quad (3.13)$$

The segment effect is expected to be explainable through proxy features ($P_{tot}, P_{sum}, P_{ins}, P_{pre}$) extracted from estimation rationale B. Adding proxy variable P_{tot} to the baseline (3.12) (Appendix A.1.4):

$$\varepsilon_t^{occ} \sim T_t + S + P_{tot}, \quad (3.14)$$

Alternatively, one can also consider the interaction between proxy and segment (Appendix B.3):

$$\varepsilon_t^{occ} \sim P_{tot} * S. \quad (3.15)$$

4 Definition of Reference Values and Errors

4.1 Reference Values as Condition-Dependent Quantities

The central challenge in productivity evaluation is that “ground-truth time-savings” is not a universal constant. The time reduction actually obtained depends on workflow conditions such as review stringency, frequency of exception handling, cross-stakeholder alignment, tool environment, and risk tolerance. Therefore, we treat reference values not as absolute ground truth, but as conditional reference estimates. In this study, the following two types of reference sources are used:

B: BM (benchmark experimental reference value): A reference value experimentally derived at the occupation level under specific evaluation procedures.

R: ref4 (log-derived reference value): A reference value estimated from interaction logs, based on different procedures and assumptions.

The main analyses in this paper use BM for task-truth decomposition (Section 5) and task-level error evaluation. ref4 is used primarily for discussion of segment trends and the semantics of reference values.

4.2 Error Based on Occupation-Truth

When task-level reference values are unavailable, a commonly used approach is to compare estimates for each task against occupation-level reference values. We define task-level error based on occupation-truth (specifically BM) as follows (taking $*$ = BM in (3.8)):

$$\varepsilon_t^{occ} = \hat{r}_t^{occ} - r_{occ}^{BM}. \quad (4.1)$$

Here r_{occ}^{BM} is the BM reference time reduction rate for occupation *occ*. This definition is simple, but there is a risk of mixing task-level and occupation-level semantics. That is, since task-level predictions are compared against occupation-level reference values, there is a risk of mixed artifacts arising when task composition ratios differ across occupations. In this paper, occupation-truth-based regression analyses are presented in A to reinforce claims about robustness and to provide contrast with the task-truth-based evaluation.

4.3 Error Based on Task-Truth (Primary Metric)

To obtain a more interpretable task-level evaluation, task-truth-based error is defined as:

$$\varepsilon_t^* = \hat{r}_t^{occ} - r_t^*. \quad (4.2)$$

Here r_t^* is the task-specific reference value estimated by the task-truth decomposition (Section 5). The sign of ε_t^* is interpreted as follows:

$\varepsilon_t^* < 0$: Conservative estimation (predicting lower time reduction than the task-truth)

$\varepsilon_t^* > 0$: Over-estimation (predicting higher time reduction than the task-truth)

5 Task-Truth Decomposition

5.1 Motivation

While occupation-level reference values may be obtained from workplace experiments or logs, task-level reference values are rarely directly observed across a wide range of occupations, even when measured for certain limited occupations and tasks. Therefore, comparing $\hat{r}_t^{occ}$ (estimated task-level time reduction rate) against the occupation-level reference value r_{occ}^{BM} can lead to mixing artifacts of task and occupation contributions. To enable a more interpretable task-level evaluation, we estimate task-specific reference values r_t^* from occupation-level reference values under a linear mixing assumption. The problem becomes particularly pronounced when task composition ratios differ across occupations. To address this, we use a minimal linear model to estimate task-level reference values r_t^* from occupation-level reference values r_{occ}^{BM} .

5.2 Linear Mixing Model

Let w_{occ}^t denote the time-share weight of occupation *occ* for task type ($t \in$ summary, instruction, presentation). The following relationship is assumed:

$$r_{occ}^{BM} \approx \sum_t w_{occ}^t r_t^*. \quad (5.1)$$

That is, the occupation-level time reduction rate is approximated by a weighted average of task-specific true values r_t^* . Since our dataset contains 11 occupations and there are only 3 unknown task-truth values r_t^* , the latter are estimated via least squares.

We report both:

(a). Unconstrained solution

(b). Solution clipped to $[0, 1]$

for the downstream analyses [file1], to avoid invalid time reduction rates (below 0 or above 1), the clipped solution is used. Under the $C[B, time]$ specification, the clipped task-truth values are:

$$r_1^* = 0.8408 \quad (\text{for summary}) \quad (5.2)$$

$$r_2^* = 0.5127 \quad (\text{for instruction}) \quad (5.3)$$

$$r_3^* = 0.4983 \quad (\text{for presentation}). \quad (5.4)$$

5.3 Decomposition Quality and Limitations

Since this decomposition is only an approximation, we evaluate the reconstruction accuracy (MAE / R^2) by comparing the reconstructed occupation-level values $\hat{r}_{occ}^{BM}$ against the actual r_{occ}^{BM} . These diagnostic results are reported in Appendix A.2.2 [file2,3]. This makes clear that task-truth values are not “observed ground truth” but rather inferred reference values. Nevertheless, they are useful in providing a more valid task-level error definition than comparing against occupation-level reference values.

(output data)

file1:[3,1] task_truth_decomposition.csv

file2:[3,1] task_truth_reconstruction_detail.csv

file3:[3,1] task_truth_reconstruction_metrics.csv

6 Results

6.1 R1. Even with task-truth-based evaluation, differences in estimation style (A/B/C) are dominant

Task-level error based on the $C[B, time]$ task-truth specification is defined as ε_t^* , and the systematic difference in error across segments (A/B/C) was first evaluated:

$$\varepsilon_t^* \sim S + T_t, \quad (6.1)$$

In this regression, both segment B and segment C showed significantly smaller (more negative) error compared to the reference segment A, confirming a tendency toward more conservative estimation than group A (coefficients: $S.B < 0$, $S.C < 0$, both significant). (See Table 2.) This segment difference (B/C being more conservative than A) is observed in the same direction in the occupation-truth-based analysis (Appendix A.1.2: see Table 5), and remains significant even after switching to the task-truth basis. This demonstrates that the error structure depends strongly not only on model capability but also on style (a policy

Table 2: Task-truth regression: $\varepsilon_t^* \sim S + T_t$

term	coefficient	p
intercept	0.2818	0
$S.B$	-0.1319	0
$S.C$	-0.2297	0
$T.pres$	-0.0162	0.2595
$T.sum$	-0.2629	0

of erring on the side of caution). Distribution Figure 2 and task-level distribution Figure 3 visually show that the positions of the error distributions shift systematically across segments. At the same time, differences by task type were also confirmed. With instruction as the reference, summary tended to be more negatively biased, with estimated reduction rates coming out lower than the task-truth. In contrast, presentation showed a small significant difference from instruction (Table 2). This task difference serves as foundational information for interpreting the importance effect described later — specifically, which tasks tend to show systematic errors.

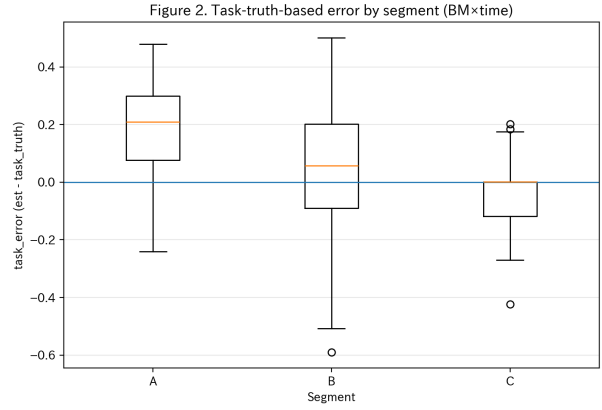


Figure 2: Task-truth-based error by segment ($C[B, time]$)

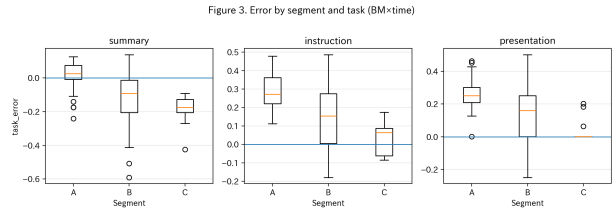


Figure 3: Error by segment and task ($C[B, time]$)

(output data)

Table 2: [4,1] TASKTRUTH_reg_segment_task.csv

Figures 2, 3: [3,2] out_Figure_2_3

6.2 R2. The rationale-based proxy is consistent with conservatism and persists even after controlling for task and segment

Next, from the rationale text provided by the model, “residual work (review/adjustment/verification, etc.)” (P_{re}), “scope broadening (from first draft to deployment/operations, etc.)” (P_{sc}), and “cautious stance (explicit references to uncertainty, assumptions, risk, and remaining human work)” (P_{ca}) were quantified using keyword matching, and their average P_{tot} was assigned as a proxy variable per model. In the main regression

$$\varepsilon_t^* \sim P_{tot} + T_t + S \quad (6.2)$$

Table 3 shows that both segment B and segment C are significantly negative relative to the reference segment A, indicating a more conservative estimation style than group A. Furthermore, P_{tot} is also significantly negative, confirming that rationale containing more residual work, cautious stance, and scope broadening is associated with more conservative estimation (suppression of over-estimation). Therefore, conservatism is explained by both segment differences and the proxy, not by either alone. The important point is that the effect of P_{tot} persists even after controlling for task differences and segment differences. That is, not merely “which group (A/B/C) a model belongs to,” but the strength of residual work and cautious stance expressed in the rationale text is consistent with the conservatism of the estimate.

However, for group B (legacy models), rationale tends to be short and P_{tot} is prone to being missing. To confirm that missing values do not distort the conclusion, sensitivity analyses were conducted under three specifications: (i) complete-case, (ii) missing=0 + missing dummy, and (iii) missing=segment mean + missing dummy (Appendix B.2). We show that the sign of the proxy effect does not reverse substantially across specifications and that the missing dummy should be interpreted as a separate factor representing “explanation style (verbosity),” thereby reinforcing the robustness of the proxy results (Appendix Table 17).

(output data):

Table 3: [4,1] TASKTRUTH_reg_proxy_task_segment.csv

Appendix Table 17: [4,2] proxy_sensitivity_comparison.csv

Table 3: Task-truth regression: $\varepsilon_t^* \sim P_{tot} + T_t + S$.

term	coefficient	p
intercept	0.4702	0
P_{tot}	-0.0277	0
$S.B$	-0.2495	0
$S.C$	-0.1517	0
$T.pres$	-0.0162	0.2458
$T.sum$	-0.2629	0

6.3 R3. The importance-task interaction is not strongly apparent in the task-truth-based evaluation

Finally, we verified whether the interaction between importance W_t and task type T systematically explains the error:

$$\varepsilon_t^* \sim W_t * T_t + S \quad (6.3)$$

In Table 4, neither the main effect of importance W_t nor the importance-task interaction $W_t : T_t$ was statistically strongly demonstrated. Therefore, one should be cautious about making a general claim that importance suppresses or amplifies errors. (In this regression too, the segment difference itself is strong, with B: -0.1504 and C: -0.2297, both significantly negative.)

This result suggests that importance patterns observable in the occupation-truth-based framework (task estimate minus occupation reference value) may have originated from mixing effects due to task-level ground truth being unobserved (i.e., treating a linear mixture of tasks as the occupation-truth). Importance still has discussion value as a proxy for “the operational conditions reflected in the reference values,” but in this dataset, segment differences and the rationale proxy have greater explanatory power as primary factors.

Table 4: Task-truth regression: $\varepsilon_t^* \sim W_t * T_t + S$.

term	coefficient	p
intercept	0.3116	0
$S.B$	-0.1504	0
$S.C$	-0.2297	0
$T.sum$	-0.2664	0
$T.pres$	-0.0035	0.9386
W_t	-0.0352	0.4301
$W_t:T.sum$	0.0007	0.9891
$W_t:T.pres$	-0.0225	0.7019

Table 4 (output data):

[4,1] TASKTRUTH_reg_importance_task_segment.csv

7 Discussion

This study showed that a perspective different from general capability metrics is important for LLM-based productivity estimation. First, models fall into estimation styles (A/B/C) that determine the “average position” of estimation error, and this can explain much of the error. In practice, rather than the expectation that “higher-performance models are always more accurate,” model selection should explicitly treat estimation policy as an axis of comparison — specifically, “how conservatively does the model estimate?” and “how far does it include residual work?”

Second, the result that proxies extracted from rationale text are consistent with conservatism suggests that time reduction rate estimation is not mere guesswork but may, at least in part, be supported by explainable factors of “residual work, scope, and cautious stance.” This leads not to blindly trusting model-provided explanations, but to an evaluation framework that quantifies explanations to compare “which model is estimating under which conditions.” In particular, the fact that proxies are more likely to be missing in group B (legacy models) should be treated as a difference in explanation style (verbosity) rather than a capability difference, making sensitivity analysis with explicit missingness markers important.

Third, the task-truth decomposition enables task-level error evaluation even when task-truth cannot be directly observed, avoiding over-claiming interactions such as importance $\times$ task. The decomposition is an approximation and does not reduce reconstruction error to zero, but it at least enables more consistent discussion than “equating occupation-truth with task-truth.” In the future, by explicitly assigning the conditions of actual operations (review intensity, rework rate, data constraints, etc.) to the reference value side as well, one could more rigorously define the “character” of the reference values and verify their correspondence with model estimation styles.

In sum, the bottleneck in LLM productivity evaluation lies not in “how smart the model is” per se, but in the inability to distinguish “what the reference values represent” from “what conditions the model assumes when estimating.” The framework proposed in this study (segmentation, rationale proxies, task-truth decomposition, and missingness sensitivity analysis) provides practical guidance for improving transparency in model selection and evaluation design for enterprise deployment.

8 Limitations

1. Reference values and condition dependence

The reference values based on BM depend on specific ex-

perimental conditions and operational assumptions. The actual time reduction in real business depends on factors such as workflow design, review stringency, frequency of exception handling, and risk tolerance. Therefore, the conclusions of this study should be interpreted as properties of the reference setting evaluated, not as “universal truths about time reduction.”

2. Task-truth decomposition is an approximation

Task-truth is estimated from occupation-level reference values using a linear mixing assumption and least-squares estimation (with values clipped to $[0, 1]$). This improves interpretability over task-occupation comparisons, but remains an estimated reference value. The decomposition fit metrics (MAE/R^2) are reported precisely to make clear that task-truth is not an observed “true value.”

3. Proxy construction is coarse and language-dependent

The proxy constructed from rationale text depends on keyword counts normalized by text length. This approach is intentionally lightweight and transparent, but may miss semantically equivalent expressions, be sensitive to phrasing, and may confuse verbosity with content. More robust NLP features (e.g., embedding similarity to residual-work concepts) could improve measurement precision, but are outside the scope of this study.

4. Proxy missingness is not random

Legacy models tend to generate short rationale, resulting in systematic proxy missingness. Sensitivity analyses using complete-case analysis and imputation with missing indicators are conducted to address this, but causal interpretation has limitations. The missing indicator should be interpreted independently as capturing a different style/verbosity factor.

5. Model set and external validity

The evaluated model set reflects the models and configurations available at the time of this study. The A/B/C segmentation is a practical classification based on observed estimation styles, and is stable in this study’s data, but the boundaries may shift if prompt policies, tool usage, or organizational context change.

6. No causal claims are made

The analyses in this study are associative in nature. We do not claim that proxy language causes conservatism; we only show that, in the reference setting evaluated, reasoning signals are consistent with estimation behavior.

9 Conclusion

This study verified the reliability of LLM-based time reduction predictions in business workflows and identified factors that systematically shape estimation error. Using 11 occupations and three common task types, task-level estimates were evaluated based on task-truth values (summary 0.8408, instruction writing 0.5127, presentation preparation 0.4983) estimated from the occupation-level BM reference under the $C[B, time]$ specification. The conclusions obtained are the following three points.

First, models exhibit stable “estimation styles” that determine the direction of systematic bias. Compared to segment A, segments B and C consistently produced more conservative estimates, and this difference was consistently observed in the task-truth-based evaluation as well.

Second, the lightweight interpretable proxies constructed from model rationale (capturing residual work, scope broadening, and cautious stance) are consistent with conservatism even after controlling for task type and segment, and their robustness was confirmed through sensitivity analysis under missingness.

Third, the adjustment effect based on importance is weak in the task-truth evaluation, and some importance patterns observed in the occupation-level comparison are suggested to originate from mixing of reference values rather than true task-level mechanisms.

In addition to these empirical findings, this study demonstrates broader implications for enterprise evaluation. Specifically, since “ground truth” for productivity is condition-dependent and definition-sensitive, reliable evaluation requires explicitly managing the meaning of reference values, calibration style, and the treatment of residual workflow tasks — not just model performance. The combination proposed in this study — segmentation, rationale-derived proxies, task-truth decomposition, and missingness sensitivity analysis — provides a practical framework for selecting and validating LLMs in production environments where productivity matters.

A Comparison of Occupation-Truth and Task-Truth

This appendix presents: (i) regression analyses that evaluate task-level predictions against occupation-truth reference values, (ii) details of the task-truth decomposition used in the main text.

These materials serve two purposes: (a) To verify that the main conclusions (e.g., differences in estimation style across segments) hold under both evaluation methods. (b) To clarify where conclusions differ when switching from the occupation-

truth to the task-truth framework (e.g., the effect of task importance).

A.1 Baseline Regression Based on Occupation-Truth

A.1.1 Error Definition (occupation-truth)

In the baseline using occupation-truth (BM unless otherwise noted), the task-level error is defined as:

$$\varepsilon_t^{occ} = \hat{r}_t^{occ} - r_{occ}^{BM}. \quad (A.1)$$

where $\hat{r}_t^{occ}$ is the model-estimated time reduction rate per task type, r_{occ}^{BM} is the BM reference value for the corresponding occupation. This baseline is a commonly used approach when task-level ground truth is unavailable. However, since task-level predictions are compared against occupation-level reference values, “mixing artifacts” may arise when task composition ratios differ across occupations.

A.1.2 Controlling for Segment and Task

We estimate the following baseline regression model:

$$\varepsilon_t^{occ} \sim S + T_t. \quad (A.2)$$

OLS (ordinary least squares) is used for estimation, with HC3 robust standard errors. The corresponding coefficient table is presented in Appendix Table 5. This table can be directly compared with the main task-truth (task-truth) regression (Figure 2, Figure 3, and segment regression Table 2). In this study’s data, the direction of estimation style differences across segments (segments B/C more conservative than segment A) is consistently observed under both occupation-truth and task-truth evaluation methods.

Table 5: Baseline regression: $\varepsilon_t^{occ} \sim S + T_t$

term	coefficient	p
intercept	0.1944	0
<i>S.B</i>	−0.1319	0
<i>S.C</i>	−0.2297	0
<i>T.pres</i>	−0.0305	0.0686
<i>T.sum</i>	0.0652	0

Table 5 (output data):

[5,1] APPX_OCCBM_reg_segment_task.csv

A.1.3 Importance–Task (occupation-truth)

The following regression model is also estimated:

$$\varepsilon_t^{occ} \sim W_t * T_t + S. \quad (A.3)$$

The results are presented in Appendix Table 6. This baseline analysis is used to contextualize the result shown in the main text that “the effect of importance is weak in task-truth-based evaluation.” For clear comparison, Appendix Table 7 presents the regression coefficients for importance under task-truth and occupation-truth (BM) side by side.

Table 6: Baseline regression: $\varepsilon_t^{occ} \sim W_t * T_t + S$

term	coefficient	p
intercept	0.2908	0
<i>S.B</i>	−0.1941	0
<i>S.C</i>	−0.2297	0
<i>T.sum</i>	−0.101	0.0206
<i>T.pres</i>	−0.0304	0.5443
<i>W_t</i>	−0.1128	0.0436
<i>W_t:T.sum</i>	−0.084	0.1842
<i>W_t:T.pres</i>	−0.0053	0.9388

Table 7: task-truth vs occ-truth: $\varepsilon_t^{occ} \sim W_t * T_t + S$

term	coef_occ	p_occ	coef_task	p_task
<i>W_t</i>	−0.1128	0.0436	−0.0352	0.4301
<i>W_t:T.sum</i>	−0.084	0.1842	0.0007	0.9891
<i>W_t:T.pres</i>	−0.0053	0.9388	−0.0225	0.7019

Results of the regression (3.7) without the segment term, $\varepsilon_t^{occ} \sim W_t * T_t$, are shown in Table 8.

Table 8: Importance–task interaction

baseline	term	slope	p	se(HC3)
BM	sum	0.0199	0.6918	0.0503
BM	inst	0.1115	0.2037	0.0721
BM	pres	0.0918	0.3427	0.0757
ref4	sum	0.1795	0.0003	0.0497
ref4	inst	0.2786	0.1788	0.0737
ref4	pres	0.0372	0.0821	0.0819

Table 6 (output data):

[5,1] APPX_OCCBM_reg_importance_task_segment.csv

Table 7 (output data):

[5,2] task_vs_occ_importance_comparison.csv

Interaction

Table 8 (output data)

[1,1] regression_coef_importance_task_BM_ref4.csv

[1,1] regression_slopes_by_task_BM_ref4.csv

A.1.4 Proxy Regression (occupation-truth)

Finally, the following proxy regression is estimated based on the occupation-truth error definition:

$$\varepsilon_t^{occ} \sim P_{tot} + T_t + S. \quad (A.4)$$

Since proxy variables are missing for some models, the complete-case results are reported here, and the related sensitivity analysis is shown in Appendix B. The results of the proxy regression based on occupation-truth are presented in Appendix Table 9, and the corresponding occupation-truth-based regression is also shown in the appendix for comparison with the main proxy regression (Table 3). Differences between occupation-truth and task-truth proxy regression results can arise from both (i) whether task estimates are compared against occupation-level or task-level reference values (error definition), and (ii) differences in the number of valid observations included in the regression due to missingness in the dependent variable or merged data.

Table 9: Proxy regression: $\varepsilon_t^{occ} \sim P_{tot} + T_t + S$

term	coefficient	p
intercept	0.3823	0
<i>P_{tot}</i>	−0.0277	0
<i>S.C</i>	−0.1517	0
<i>S.B</i>	−0.2495	0
<i>T.pres</i>	−0.0305	0.0631
<i>T.sum</i>	0.0652	0

Table 9 (output data):

[5,1] APPX_OCCBM_reg_proxy_task_segment.csv

A.1.5 Triple Interaction

Furthermore, using (3.11) as the base model, the full model

$$\varepsilon_{task}^{occ} \sim W_t * T_t * S, \quad (A.5)$$

was estimated, both using HC3 robust standard errors. As a result, the full model improved fit over the base model (BM: $R^2=0.303 \rightarrow 0.340$, $AIC=-486.19 \rightarrow -493.74$; ref4: $R^2=0.242 \rightarrow 0.299$, $AIC=-312.64 \rightarrow -331.69$), confirming that explicitly accounting for segment differences is effective in explaining the error structure (Table 10). Table 10 (output data): [1.2] segmented_regression_fit.csv

Table 10: Effect of segment introduction: R^2 /AIC

baseline	spec	R^2	AIC
BM	base	0.30273	-486.187
BM	full	0.34048	-493.742
ref4	base	0.24188	-312.643
ref4	full	0.29939	-331.693

Regarding the mean bias (intercept difference) per segment, in the base model, both B and C were significantly shifted toward lower error (more under-estimation side) relative to A as reference. Specifically, under BM, B showed -0.194 ($p < 10^{-16}$) and C showed -0.230 ($p < 10^{-42}$); under ref4, B showed -0.077 ($p = 0.009$) and C showed -0.230 ($p < 10^{-29}$). This suggests that most of the estimation error can be explained by “a consistent estimation bias (conservative/non-conservative) per model group, prior to importance or task effects” (Table 11). Table 11 (output data): [1,2] segmented_regression_key_terms.csv

Table 11: Regression coefficients: segment bias

baseline	spec	term	coef.	se(HC3)	p
BM	base	S.B	-0.1941	0.0229	2.6E-17
BM	base	S.C	-0.2297	0.0166	2.2E-43
ref4	base	S.B	-0.0774	0.0296	0.0091
ref4	base	S.C	-0.2297	0.0201	2.6E-30

On the other hand, regarding the segment difference in importance response for the instruction task ($W_t * T.ins * S$) (Table 12), positive coefficients were estimated in the full model but did not reach statistical significance (BM: B $+0.358$, $p=0.166$; C $+0.181$, $p=0.540$; ref4: B $+0.393$, $p=0.223$; C $+0.181$, $p=0.518$). Therefore, in this dataset, rather than claiming that heterogeneity in importance effects for instruction “clearly differs across segments,” it is more appropriate to conclude that the intercept difference (mean bias) per segment makes the dominant contribution. Table 12 (output data): [1,2] segmented_regression_key_terms.csv

Table 12: Regression coefficients $W_t * T.ins * S$

baseline	spec	term	coef.	se(HC3)	p
BM	full	$W_t:T.ins:S.B$	0.3577	0.2584	0.1663
BM	full	$W_t:T.ins:S.C$	0.1809	0.2955	0.5404
ref4	full	$W_t:T.ins:S.B$	0.3929	0.3222	0.2226
ref4	full	$W_t:T.ins:S.C$	0.1809	0.2796	0.5176

A.2 Details of Task-Truth Decomposition

A.2.1 Decomposition Results

Least squares estimation of r_t^* is performed based on (5.1) covering all occupations (11 occupations in this study).

Table 13: Estimated task-truth (bare)

baseline	weight	r_1^*	r_2^*	r_3^*
BM	time	0.8408	0.5127	0.4983
BM	imp	0.7499	0.5424	0.5483
ref4	time	0.8406	0.4309	1.1099
ref4	imp	0.8396	0.5250	1.0189

Since time reduction rates should inherently fall within $[0, 1]$, the estimated values $\hat{r}_t^*$ are clipped to $[0, 1]$, and the clipped values are used in downstream analyses.

Table 14: Estimated task-truth (clipped)

baseline	weight	r_1^*	r_2^*	r_3^*
BM	time	0.8408	0.5127	0.4983
BM	imp	0.7499	0.5424	0.5483
ref4	time	0.8406	0.4309	1
ref4	imp	0.8396	0.5250	1

The estimated task-truth (clipped) values under the $C[B, time]$ setting are:

$$r_{sum}^* = 0.8408 \quad (\text{summary}) \quad (\text{A.6})$$

$$r_{ins}^* = 0.5127 \quad (\text{instruction}) \quad (\text{A.7})$$

$$r_{pre}^* = 0.4983 \quad (\text{presentation}) \quad (\text{A.8})$$

A.2.2 Decomposition Fit and Reconstruction Error

To quantify how well the decomposition model explains the occupation-level reference values, the reconstruction values are computed as:

$$\hat{r}_{occ}^{BM} = \sum_t w_{occ}^t \hat{r}_t^* \quad (\text{A.9})$$

and these reconstructed values are compared against the actual occupation-level reference values r_{occ}^{BM} , reporting reconstruction error (MAE / RMSE / R^2) (see Appendix Figure 4, metrics Table 15, and error Table 16). These diagnostic results clearly show that task-truth values are inferred reference values, not observed “true values.” Nevertheless, they are useful in providing a more valid task-level error definition than comparing task predictions against occupation-level reference values.

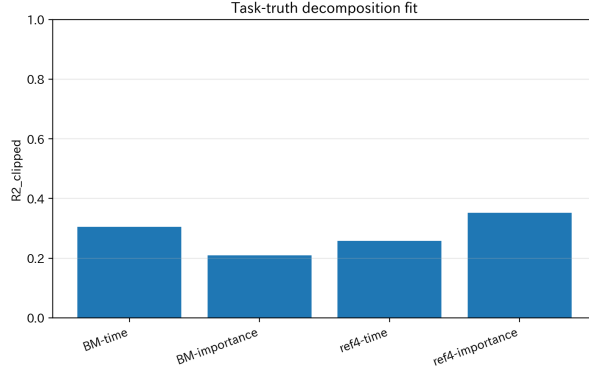


Figure 4: Task-truth decomposition fit

Table 15: Reconstruction error MAE / RMSE / R^2 (clipped)

baseline	weight	MAE	RMSE	R^2
BM	time	0.0584	0.0700	0.3047
BM	imp	0.0617	0.0746	0.2101
ref4	time	0.0914	0.1084	0.2584
ref4	imp	0.0791	0.1013	0.3520

Table 16: Reconstruction error for $C[B, time]$ occupation-truth $r_{occ}^{BM}, \hat{r}_{occ}^{BM}$

occ_id	occ_name	true value	reconst.	abs_error
1	Management Consultant	0.5423	0.5726	0.0302
2	Business Planning	0.5198	0.5547	0.0349
2.2	Marketing	0.5473	0.5868	0.0395
3	PM/PdM	0.5467	0.5412	0.0055
4	Writer/Editor	0.612	0.7067	0.0947
5	Sales Assistant	0.5735	0.6083	0.0348
6	Training Instructor	0.665	0.5697	0.0953
7	Finance/Accounting	0.8	0.6575	0.1425
7.2	Finance/Accounting 2	0.6164	0.5764	0.0400
8	Customer Support	0.68	0.6418	0.0382
8.2	Customer Support 2	0.4984	0.5850	0.0866

(output data):

Figure 4: [3,3] Figure_task_truth_decomposition_R2fit.png

Task-truth decomposition [3,1]
task_truth_decomposition.csv

Table 15 [3,1] task_truth_reconstruction_metrics.csv

Table 16 [3,1] task_truth_reconstruction_detail.csv

B Proxy Construction and Sensitivity Analysis

This appendix describes the proxy features derived from rationale text and the sensitivity analysis conducted to address systematic missingness.

B.1 Keyword Dictionary and Normalization

B.1.1 Proxy Definition

We define a model-level proxy score $P_\tau = \{P_{re}, P_{sc}, P_{ca}\}$ using three interpretable components extracted from rationale text. For each model, keyword occurrences in the extracted rationale text are counted and normalized by text length (occurrences per 1,000 characters) to obtain P_τ . The keyword components are:

1. Residual work P_{re} : Keywords indicating verification, review, approval, adjustment, exception handling, and compliance-related tasks.
2. Scope broadening P_{sc} : Keywords indicating that the estimate covers an end-to-end workflow scope (e.g., “first draft” vs. “deployment / operations”).
3. Caution / uncertainty P_{ca} : Keywords indicating explicit references to uncertainty, assumptions, risk, and remaining human tasks.

The specific keyword lists and normalization procedures are included in the publicly available pipeline code (Appendix D). P_{tot} is computed as the average of these three component scores.

B.1.2 Interpretation and Caveats

This proxy is intentionally designed to be lightweight and transparent. As such, it does not capture all semantic variations and may correlate with verbosity. As shown in Appendix B.3, the proxy alone currently has almost no standalone explanatory power.

Therefore, P_{tot} is treated as an “interpretable signal” indicating the assumptions and caution explicitly stated by the model, and not as causal evidence.

B.2 Missingness Sensitivity Analysis

P_τ tends to be missing more frequently for legacy models that generate short rationale. To ensure that conclusions about proxies do not become artifacts of the missingness handling method,

analyses are conducted under three specifications: (i) Complete-case: estimation using only models for which all P_τ values are available.

(ii) Zero-imputation + missing indicator: missing P_τ values are replaced with 0, and a binary variable indicating missingness is added.

(iii) Segment-mean imputation + missing indicator: missing P_τ values are replaced with the mean within the same segment, and a missing indicator is also added.

Coefficients and p-values are reported in Appendix Table 17. Comparing these specifications, the effect of proxy does not reverse in sign, while the missing indicator captures a different factor (e.g., differences in verbosity or style) and should be interpreted independently of P_{tot} .

That is, the conclusion that “the more residual work, cautious stance, and scope broadening signals are present in the rationale, the more conservative the estimate” does not depend on the method of handling missing values. It is also important to note that the missing indicator (proxy_miss) is not significant in either specification: 0.044 (p=0.4077) and 0.0463 (p=0.3766), meaning that “the cases where proxy is unavailable/zero” cannot be said to explain the error by themselves. In other words, the proxy effect is likely not a side effect of missingness, but corresponds to the actual content of the proxy.

Table 17: Sensitivity analysis for proxy missingness

spec	Complete only	All models: proxy0+miss.	All models: seg.mean+miss.
proxy_term	proxy_total	proxy0	proxy_segmean
proxy_coef	−0.0280	−0.0277	−0.0313
proxy_p	0.000	0.000	0.000
missing_term	-	proxy_miss	proxy_miss
missing_coef	-	0.0440	0.0463
missing_p	-	0.4077	0.3766

Table 17 (output data):

[4,2] proxy_sensitivity_comparison.csv

B.3 Explanatory Power of Proxy Variables

While the explanatory power of proxy variables is limited in the current dataset, the analyses suggest that interaction terms and feature design could potentially yield combined effects with segment variables that reduce error.

$$\varepsilon_t^{occ} \sim P_{tot} + S(A) \quad (B.1)$$

where $S(A)$ denotes segment A as the reference.

$$\varepsilon_t^{occ} \sim P_{tot} * S(A) \quad (B.2)$$

Table 18: Regression coefficients: segment addition (B.1) and interaction (B.2)

term	coef ₊	p ₊	coef _*	p _*
const	0.4007	0.0105	0.4427	0.0236
$S.B$	−0.2578	0.0139	−0.3618	0.0933
$S.C$	−0.1526	0.0704	0.4810	0.6856
P_{tot}	−0.0302	0.1357	−0.0364	0.1573
$P_{tot}:S.B$	-	-	0.0305	0.5194
$P_{tot}:S.C$	-	-	−0.0642	0.6081

Table 18 (output data):

[2,1] BM_proxy_total_segment_A.csv

Table 19: Correlation between proxy features and BM error. $P_{A,B,C}$ is P_{tot} within each segment.

proxy	n	pearson _r	pearson _p	spearman _r	spearman _p
P_A	7	−0.6394	0.1220	−0.8571	0.0137
P_B	6	−0.0648	0.9029	−0.2571	0.6228
P_C	2	-	-	-	-
P_{tot}	15	−0.0218	0.9386	−0.0964	0.7325
P_{re}	15	0.0703	0.8035	0.0107	0.9698
P_{sc}	15	−0.1295	0.6454	−0.0464	0.8695
P_{ca}	15	−0.0994	0.7246	−0.0036	0.9899

From Table 18, a tendency is visible for segment A where larger P_{tot} corresponds to smaller BM error (the P_{tot} coefficient is negative in both cases); Table 19 also shows a negative correlation for A. For segment B, the combined coefficient including the main effect is −0.0059, nearly zero, and Table 19 also shows no correlation. These suggest the possibility that the proxy operates differently depending on the segment, but the incremental effect through interaction is small ($R^2 = 0.702 \rightarrow 0.727$; without the segment term $R^2 = 0.001$; for (B.1) $R^2 = 0.702$; for (B.2) $R^2 = 0.727$). (Table 22)

(output data):

Table 19 [2,1] proxy_BM.csv; [2,1] proxy_total_vs_BM.csv

Table 22 [2,1] proxy_R2.csv

B.3.1 Centered Proxy Regression

To absorb within-segment variation of proxy features P_τ , the within-segment mean $\bar{P}_\tau$ is subtracted (centering):

$$\mathcal{P}_\tau = P_\tau - \bar{P}_\tau, \quad P_\tau \in \text{Segment} \quad (B.3)$$

Centering is applied not only to the three proxy variables P_{re} , P_{sc} , P_{ca} but also to P_{tot} , and the following are examined:

$$\varepsilon_{task}^{occ} \sim \mathcal{P}_{tot} + S(A) \quad (B.4)$$

$$\varepsilon_{task}^{occ} \sim \mathcal{P}_{re} + \mathcal{P}_{sc} + \mathcal{P}_{ca} + S(A) \quad (B.5)$$

Centering increases the absolute value of correlation coefficients and decreases p-values. While p remains high and does not reach significance, an increasing trend in explanatory power is observed (Table 20: output [2,3] centered_correlation_table.csv).

Table 20: Effect of centering: correlation table 19 for P_{tot} and P_{re}

proxy	n	pearson _r	pearson _p	spearman _r	spearman _p
P_{tot}	15	-0.0218	0.9386	-0.0964	0.7325
$\mathcal{P}_{tot}$	15	-0.265	0.3398	-0.4393	0.1014
P_{re}	15	0.0703	0.8035	0.0107	0.9698
$\mathcal{P}_{re}$	15	-0.134	0.6339	-0.2429	0.3831

Table 21: Regression coefficients: aggregate proxy (B.4) and three components (B.5)

term	coef _t	p _t	coef ₃	p ₃
const	0.1954	0	0.1954	0
$S.B$	-0.1296	0.0059	-0.1296	0.0088
$S.C$	-0.2376	0.0012	-0.2376	0.0022
$\mathcal{P}_{tot}$	-0.0302	0.1357	-	-
$\mathcal{P}_{re}$	-	-	0.0070	0.6704
$\mathcal{P}_{sc}$	-	-	-0.0065	0.8556
$\mathcal{P}_{ca}$	-	-	-0.0388	0.2963

Table 21 (output data):

[2,3] centered_regression_components.csv

[2,3] centered_regression_proxy_total.csv

Looking at Table 22, the incremental effect on explanatory power is larger by decomposing into three components than by interaction or centering ($R^2 = 0.702 \rightarrow 0.745$; without segment term $R^2 = 0.001$; for (B.4) $R^2 = 0.702$; for (B.5) $R^2 = 0.745$).

Table 22 (output data):

[2,1] proxy_R2.csv; [2,3] centered_regression_r2.csv

C Occupation Data and Model List

Table 22: Comparison of explanatory power

spec	R^2
P_{tot} only	0.0005
$P_{tot} + S$	0.7018
$\mathcal{P}_{tot} + S$	0.7018
$P_{tot} * S$	0.7265
$P_{re} + P_{sc} + P_{ca} + S$	0.745
$\mathcal{P}_{re} + \mathcal{P}_{sc} + \mathcal{P}_{ca} + S$	0.745

C.1 Occupation Data List

Table 23: Values used consistently throughout this note. Time ratios and importance for three tasks per occupation, actual time reduction rates {ref.1, ref.2, ref.4} and their average (BM). The actual values were selected from the literature[1]-[7]. An overview of the results from these measurement experiments is given in Section 0, and the method for constructing occupation-truth and task composition is described in detail in Section 1.

C.2 List of Models Used

Table 24: List of LLMs used in this note. Model names used, their IDs, and assigned segments. Performance values are linked to the corresponding model names and performance values from OpenLM Chatbot Arena. These are used to analyze the relationship between Elo and error.

C.3 Errors from Occupation-Truth

A list of errors from occupation-truth in the case where the time reduction rates computed based on LLM inference (human work time, AI work time, and their estimation rationale for three tasks) are used as LLM estimates, computed as the time-ratio-weighted average. (C.3.1)

The case using the importance-weighted time-weighted average is shown in C.3.2.

These tables exist for each model, but in this note only two representative models are shown as examples. One example from the conservative-estimating segment C group: gpt51T (GPT-5.1 Thinking), and one example from the advanced-estimating segment A group: haiku45 (Claude Haiku 4.5).

All model data is stored at:

raw data (output text from LLMs):

[2,1] in_proxy_new

Data formatted for analysis (input data):

[0] Time_Weighted_Average_Data_W.csv

[0] Importance_Weighted_Average_Data_W.csv

Table 23: Task characteristics: time ratio w_t , importance W_t , ground-truth time reduction rates. For occupations 7 and 8 in ref.2, values from [3] and [7] are used.

Occupation		1. Summarization		2. Instruction Writing		3. Presentation		Time Reduction Rate (Actual)			Avg
		Time ratio	Importance	Time ratio	Importance	Time ratio	Importance	ref.1	ref.2	ref.4	
1	Management Consultant	20%	0.7	40%	0.9	40%	0.9	42.2%	37%	83.5%	54%
2	Business Planning	15%	0.6	35%	0.8	50%	0.9	38.95%	37%	80%	52%
2.2	Marketing	25%	0.7	20%	0.6	55%	0.8	40.2%	37%	87%	55%
3	Project Manager	10%	0.5	60%	0.9	30%	0.7	42%	37%	85%	55%
4	Editor/Writer	60%	0.9	20%	0.5	20%	0.4	55.6%	37%	91%	61%
5	Sales Assistant/Secretary	30%	0.6	50%	0.8	20%	0.5	48.7%	-	66%	57%
6	Training Instructor	20%	0.5	20%	0.6	60%	0.9	38%	-	95%	67%
7	Finance/Accounting	45%	0.9	35%	1	20%	0.95	-	80%	80%	80%
7.2	Finance/Accounting 2	21%	0.82	43%	0.92	36%	0.88	43.3%	-	80%	62%
8	Customer Support	40%	1	45%	0.95	15%	0.8	-	80%	56%	68%
8.2	Customer Support 2	22%	0.72	44%	0.9	35%	0.84	43.7%	-	56%	50%

Table 24: Correspondence table of models used and Chatbot Arena metrics

model id	Model used	segment	OpenLM model	Arena Elo	AAII	MMLU-Pro
gpt51T	GPT-5.1 Thinking	C	GPT-5.1-high	1464	70	87.1
gpt52P	GPT-5.2 Pro	C	GPT-5.2-high	1465	72	87.4
gpt52I	GPT-5.2 Instant	A	GPT-5.2	1464	68	87.4
gpt52T	GPT-5.2 Thinking	A	GPT-5.2-high	1465	72	87.4
sonnet46	Sonnet 4.6	A	Claude Sonnet 4.6	1446	65	87.3
sonnet46T	Sonnet4.6 Ext. Thinking	A	Claude Sonnet 4.6	1446	65	87.3
opus46	Opus 4.6	A	Claude Opus 4.6	1490	71	89.5
opus46T	Opus 4.6 Ext. Thinking	A	Claude Opus 4.6 Thinking	1503	73	89.7
haiku45	Haiku 4.5	A	Claude Haiku 4.5	1378	42	80
gpt35t	GPT-3.5-turbo-0125	B	GPT-3.5-turbo-0613	1145	9	46.2
gpt4om	GPT-4o-mini-2024-07-18	B	GPT-4o-mini-2024-07-18	1289	22	64.8
gem2:9b	Gemma2-9b	B	Gemma2-9b-it	1213	8	49.5
llama32:3b	Llama3.2-3b	B	Llama3.2-3b-instruct	1118	5	34.7
mist:7b	Mistral-7b-ins-v02	B	Mistral-7b-inst-v02	1097	1	24.5
llama2:13b	Llama2-13b-chat	B	Llama2-13b-chat	1093	6	40.6

C.3.1 Time-weighted average case

Table 25 shows one example from the conservative segment C group: gpt51T (GPT-5.1 Thinking), and Table 26 shows one example from segment A group: haiku45 (Claude Haiku 4.5).

C.3.2 Importance-weighted case

Table 27 shows GPT-5.1 Thinking and Table 28 shows Haiku 4.5, with importance weights applied.

The importance weight acts as a factor that reduces the estimated time reduction rate, thereby also reducing the BM reference error for groups A and B.

D Data and Code

To ensure reproducibility, the CSV input files, Python code, and output results are provided at the following URL: https://github.com/pub5023/productivity_indicator

Folder structure:

- 0 input data: Arena Elo data, time estimation data per model, w_t , W_t data
- 1 occ_truth_regression
 - 1.1 importance_regression: regression analysis of Elo vs. error ($imp * T$), model scatter plots
 - 1.2 segmented_regression: segment interaction ($imp * T * S$, $S + T$)
- 2 proxy_analysis
 - 2.1 proxy_preparation: keyword dictionary, rationale data, proxy feature computation
 - 2.2 proxy_regression: proxy regression analysis ($P * S$, $P + S$), centered regression
- 3 task_truth_decomposition
- 4 task_truth_regression: regression ($S + T$, $P + T + S$, $imp * T + S$), proxy sensitivity analysis
- 5 appendix_occ_truth_results: occupation-truth-based regression ($S + T$, $P + T + S$, $imp * T + S$)

Table 25: Deviation and divergence rate of estimated time reduction rates (GPT-5.1 Thinking)

	Occupation (occ)	Reduction 1	Reduction 2	Reduction 3	Weighted avg	BM	ref.4	ε_1	χ_1	ε_2	χ_2
1	Management Consultant	75%	60%	50%	59%	54%	84%	5%	9%	-25%	-29%
2	Business Planning	75%	63%	50%	60%	52%	80%	8%	15%	-20%	-25%
2.2	Marketing	75%	69%	68%	70%	55%	87%	15%	27%	-17%	-20%
3	Project Manager	75%	57%	50%	58%	55%	85%	3%	6%	-27%	-32%
4	Editor/Writer	71%	61%	50%	59%	61%	91%	-2%	-4%	-32%	-35%
5	Sales Assistant/Secretary	75%	64%	50%	61%	57%	66%	3%	6%	-5%	-8%
6	Training Instructor	71%	58%	50%	58%	67%	95%	-9%	-13%	-37%	-39%
7	Finance/Accounting	67%	58%	50%	57%	80%	80%	-23%	-29%	-23%	-29%
7.2	Finance/Accounting 2	67%	58%	50%	57%	62%	80%	-5%	-8%	-23%	-29%
8	Customer Support	60%	60%	70%	64%	68%	56%	-4%	-6%	8%	14%
8.2	Customer Support 2	70%	60%	56%	61%	50%	56%	11%	21%	4%	8%

Table 26: Deviation and divergence rate of estimated time reduction rates (Haiku 4.5)

	Occupation (occ)	Reduction 1	Reduction 2	Reduction 3	Weighted avg	BM	ref.4	ε_1	χ_1	ε_2	χ_2
1	Management Consultant	93%	85%	80%	84%	54%	84%	30%	56%	0%	0%
2	Business Planning	93%	88%	80%	85%	52%	80%	33%	63%	5%	6%
2.2	Marketing	95%	90%	90%	91%	55%	87%	36%	65%	4%	5%
3	Project Manager	95%	90%	95%	91%	55%	85%	36%	66%	6%	7%
4	Editor/Writer	92%	90%	90%	91%	61%	91%	30%	49%	0%	0%
5	Sales Assistant/Secretary	95%	90%	95%	92%	57%	66%	35%	61%	26%	39%
6	Training Instructor	95%	92%	95%	94%	67%	95%	27%	40%	-1%	-1%
7	Finance/Accounting	95%	98%	95%	96%	80%	80%	16%	20%	16%	20%
7.2	Finance/Accounting 2	95%	95%	95%	95%	62%	80%	33%	53%	15%	19%
8	Customer Support	95%	95%	90%	94%	68%	56%	26%	38%	38%	68%
8.2	Customer Support 2	90%	95%	90%	92%	50%	56%	42%	84%	36%	64%

Table 27: Deviation and divergence rate of estimated time reduction rates with importance weighting (GPT-5.1 Thinking)

	Occupation (occ)	Reduction 1	Reduction 2	Reduction 3	Weighted avg	BM	ref.4	ε_1	χ_1	ε_2	χ_2
1	Management Consultant	75%	60%	50%	63%	54%	84%	9%	17%	-21%	-25%
2	Business Planning	75%	63%	50%	61%	52%	80%	9%	17%	-19%	-24%
2.2	Marketing	75%	69%	68%	70%	55%	87%	15%	27%	-17%	-20%
3	Project Manager	75%	57%	50%	59%	55%	85%	4%	7%	-26%	-31%
4	Editor/Writer	71%	61%	50%	67%	61%	91%	6%	10%	-24%	-26%
5	Sales Assistant/Secretary	75%	64%	50%	63%	57%	66%	6%	11%	-3%	-5%
6	Training Instructor	71%	58%	50%	56%	67%	95%	-11%	-16%	-39%	-41%
7	Finance/Accounting	67%	58%	50%	60%	80%	80%	-20%	-25%	-20%	-25%
7.2	Finance/Accounting 2	67%	58%	50%	58%	62%	80%	-4%	-6%	-22%	-28%
8	Customer Support	60%	60%	70%	61%	68%	56%	-7%	-10%	5%	9%
8.2	Customer Support 2	70%	60%	56%	62%	50%	56%	12%	24%	6%	11%

Table 28: Deviation and divergence rate of estimated time reduction rates with importance weighting (Haiku 4.5)

Occupation (occ)	Reduction 1	Reduction 2	Reduction 3	Weighted avg	BM	ref.4	ε_1	χ_1	ε_2	χ_2
1 Management Consultant	93%	85%	80%	86%	54%	84%	32%	59%	2%	2%
2 Business Planning	93%	88%	80%	86%	52%	80%	34%	65%	6%	8%
2.2 Marketing	95%	90%	90%	91%	55%	87%	36%	65%	4%	5%
3 Project Manager	95%	90%	95%	92%	55%	85%	37%	67%	7%	8%
4 Editor/Writer	92%	90%	90%	92%	61%	91%	31%	51%	1%	1%
5 Sales Assistant/Secretary	95%	90%	95%	92%	57%	66%	35%	61%	26%	39%
6 Training Instructor	95%	92%	95%	94%	67%	95%	27%	40%	-1%	-1%
7 Finance/Accounting	95%	98%	95%	96%	80%	80%	16%	20%	16%	20%
7.2 Finance/Accounting 2	95%	95%	95%	95%	62%	80%	33%	53%	15%	19%
8 Customer Support	95%	95%	90%	95%	68%	56%	27%	40%	39%	70%
8.2 Customer Support 2	90%	95%	90%	92%	50%	56%	42%	84%	36%	64%

Table 29: Model comparison: mean values of time reduction rate, deviation, and divergence rate

model	Time reduction rate				BM		ref.4	
	Summary	Instruction	Presentation	Time-weighted	ε_1	χ_1	ε_2	χ_2
GPT-5.1 Thinking	71%	61%	54%	60%	8%	13%	20%	25%
GPT-5.2 Pro	63%	47%	50%	51%	9%	13%	27%	32%
GPT-5.2 Instant	85%	75%	75%	77%	18%	32%	10%	13%
GPT-5.2 Thinking	84%	74%	74%	76%	17%	30%	10%	14%
Sonnet 4.6	87%	82%	75%	80%	18%	32%	10%	13%
Sonnet4.6 Ext. Thinking	90%	87%	80%	85%	25%	44%	9%	14%
Opus 4.6	78%	70%	69%	71%	13%	23%	13%	17%
Opus 4.6 Ext. Thinking	83%	76%	70%	75%	16%	29%	8%	10%
Haiku 4.5	94%	92%	92%	93%	33%	57%	15%	23%
GPT-3.5-turbo-0125	65%	64%	63%	64%	14%	24%	24%	31%
GPT-4o-mini-2024-07-18	80%	73%	70%	73%	14%	26%	12%	17%
Gemma2-9b	68%	61%	61%	63%	8%	14%	21%	26%
Llama3.2-3b	65%	63%	50%	58%	10%	17%	27%	33%
Mistral-7b-ins-v02	81%	80%	76%	79%	20%	35%	16%	22%
Llama2-13b-chat	68%	60%	55%	60%	21%	36%	25%	33%

Table 30: Model comparison: mean values of time reduction rate, deviation, and divergence rate (importance-weighted)

model	Time reduction rate				BM		ref.4	
	Summary	Instruction	Presentation	Time-weighted	ε_1	χ_1	ε_2	χ_2
GPT-5.1 Thinking	51%	49%	42%	47%	14%	22%	32%	38%
GPT-5.2 Pro	45%	39%	39%	40%	20%	33%	38%	47%
GPT-5.2 Instant	61%	61%	58%	60%	10%	16%	22%	27%
GPT-5.2 Thinking	61%	60%	57%	59%	9%	15%	23%	28%
Sonnet 4.6	62%	66%	58%	62%	10%	18%	19%	23%
Sonnet4.6 Ext. Thinking	65%	70%	62%	66%	10%	19%	17%	21%
Opus 4.6	56%	57%	54%	55%	9%	14%	24%	29%
Opus 4.6 Ext. Thinking	59%	61%	55%	58%	9%	15%	22%	26%
Haiku 4.5	68%	74%	72%	72%	14%	25%	17%	23%
GPT-3.5-turbo-0125	46%	52%	49%	50%	13%	22%	32%	38%
GPT-4o-mini-2024-07-18	58%	59%	56%	57%	10%	17%	25%	30%
Gemma2-9b	48%	50%	49%	49%	12%	21%	31%	37%
Llama3.2-3b	49%	51%	40%	46%	17%	28%	33%	41%
Mistral-7b-ins-v02	59%	66%	60%	62%	14%	24%	24%	30%
Llama2-13b-chat	48%	48%	42%	46%	21%	36%	33%	41%

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